

Scaling Education Reform: Evidence from Morocco’s Pioneer School Program

Year 3 Preliminary Results

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1 Introduction

This report presents preliminary estimates of the impact of Morocco’s Pioneer School Program (PSP) after three years. The PSP was implemented in 626 Moroccan primary schools in September of 2023; these schools are called Cohort 1 schools. Another 2,000 primary schools implemented it in September of 2024, and another 2,000 implemented it in September of 2025; these are Cohort 2 and Cohort 3 schools, respectively. This report presents preliminary estimates of the impact of the PSP on all three cohorts of schools.

These results are preliminary, for two reasons. First, for the final report we will use Item Response Theory (IRT) to generate comparable values of students skills for each of the three subjects (Arabic, French and mathematics) for all assessments (tests) taken over the three years and for all grades. This will allow us to standardize all assessment results using the same distribution of test scores; we were unable to apply IRT analysis in this report due to time constraints. In this preliminary report, each test is standardized using the comparison group for each exam, which means that, for example, the grade 2 exam scores (number of correctly answered questions) at endline for any given cohort in any given year are standardized using the distribution of the grade 2 endline exam among the matched comparison schools for that cohort in that year. This means that the standardized estimated impacts in this report are not directly comparable to the standardized estimated impacts in previous final reports, although in most cases they are similar.

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The second reason why this report is preliminary is that we have not had time to fully clean the data and check our analysis. However, we have done extensive cleaning and checking, and so we expect that these results will be similar to those in the final report (but we cannot guarantee this).

The new results in this report can be summarized as follows. First, for Cohort 1, after three years of exposure the estimated impacts are somewhat smaller than the estimated impacts after two years of exposure (and both of those impacts are much smaller than the estimated impacts after one year of exposure). Second, for Cohort 2, the estimated impacts after two years of exposure are, on average, somewhat higher than the estimated impacts after one year of exposure, and all of this increase is due to the higher estimated impact in mathematics. Third, for Cohort 3, the estimated impacts after one year of exposure are somewhat smaller than the estimated impacts for Cohort 2 after one year of exposure, but they are still highly significant and larger than most of the estimated impacts of education programs implemented in other developing countries.

To complete the Year 3 final report for the roll out of PSP, there are several data sets that we require. For each data set, we provide a justification.

- *Dropout data:* This is a secondary outcome of interest in our analysis.
- *Grade 6 exams:* A major motivation of this analysis is the success of primary school students to the promotion of secondary school. We would like to evaluate the impact of PSP on primary school graduation and secondary school acceptance.
- *Student baseline characteristics for Cohort 2 and 3:* As part of our LASSO control design and heterogeneity analysis, we are missing some important student level characteristics for cohort 2 and 3. This will improve the precision of our estimates and expand the exploration of our analysis.
- *Implementation data:* For all three years of PSP, we do not have any validation of the fidelity of implementation. This will inform us how the program can be improved as well as highlight strengths in terms of implementation.
- *Student and teacher movement between schools:* One potential explanation for our results is through a spillover channel. To test this channel, we need information on which PSP and comparison school students and teachers move to other schools.
- *Financial costs of PSP:* We would like to perform a cost-effectiveness analysis with the expansion of PSP. To do this, we need information on the capacity and cost of implementing the four main components of PSP.

2 Preliminary Results

This section reports preliminary results of the impact of the PSP after three years. More specifically, it presents estimates of the intent-to-treat effect of the PSP intervention on student learning using a strategy that combines difference-in-differences estimation with matching

methods. This strategy compares changes over time in the outcome variables of interest for primary schools that were assigned to receive that intervention with the same changes for their matched comparison schools. For all outcomes, the following empirical specification is used:

$$Y_{igsjt} = \beta(TREAT_{sj} \times POST_t) + \delta(X_{igsj} \times POST_t) + \zeta(\eta_j \times \theta_g \times POST_t) + \gamma_{igsj} + \varepsilon_{igsjt} \quad (1)$$

Here, Y_{igsjt} is outcome Y for student i in grade g in school s in matched pair j in period t . $TREAT_{sj}$ is the assignment of school s in matched pair j to receive the education intervention, and $POST_t$ is a dummy variable equal to 1 if the outcome Y is measured at endline (i.e., after the PSP was rolled out) and 0 if measured at baseline (i.e., before the PSP was rolled out). This is the canonical difference-in-differences design.

The $\gamma_{i,g,s,j}$ term is a student fixed effect that measures the impacts on Y_{igsjt} of all student characteristics, observed or unobserved, that do not change over time. However, the impacts of some student characteristics may change over time, so a vector of observed student characteristics, denoted by X_{igsj} , is added and multiplied by $POST_t$. The corresponding vector of coefficients for this interaction term, denoted by δ , measures the change in the impacts of these observed student characteristics over time. In other words, δ measures the contributions of observed student characteristics to changes in the outcome variables over time. Since nearly all students stay in the same school between baseline and endline data collection, student fixed effects also include school fixed effects, and so the X_{igsj} variables can also include observed school variables. The student variables available in the dataset are students' gender and whether the student was eligible for a social safety net program (measured at baseline). The school characteristics are school-level average test scores, region, average class size, and the student-to-teacher ratio (all measured at baseline). A LASSO (Least Absolute Shrinkage and Selection Operator) procedure was used to determine which $X_{igsj} \times POST_t$ interactions to include in the specification.¹ The three-way interaction $\zeta(\eta_j \times \theta_g \times POST_t)$ represents grade-by-matched pair interactions with the post-period indicator. Intuitively, these fixed effects account for how much, on average, a PSP student's matched-pair comparison peers (who are enrolled in the same grade in the matched non-PSP schools) learned between baseline and endline; these effects also apply to corresponding PSP students, which in effect implements the parallel trends assumption.

The coefficient of interest is β ; it measures the intent-to-treat (ITT) effect of assignment to the program. In estimating ITT effects, we allow for the (plausible) possibility that the program was not fully implemented in at least some of the PSP schools (none of the non-PSP schools received the program). Following de Chaisemartin and Ramirez-Cuellar (2024) on clustering in paired and small-strata experiments, we cluster standard errors at the matched-school-pair level.

For β in Equation (1) to estimate the causal impact of the PSP program on the outcomes of interest, the identifying assumption is that, conditional on the "comparison" variables in that

¹More specifically, we first calculated each student's change in test scores between the baseline and endline assessments. Then, we residualized this change in test scores, subtracting the comparison group's grade-by-matched pair mean. Lastly, with these residuals, we use a post-double-selection LASSO to select relevant predictors X_{igsj} (Belloni et al., 2011).

Table 1: Students' Performance After 3-Year Rollout

	Cohort 1 (2023/24)				Cohort 2 (2024/25)				Cohort 3 (2025/26)				Cohorts Pooled			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
<i>Panel A: 1-Year Exposure</i>																
All students	0.903*** (0.030) [0.000]	0.489*** (0.044) [0.000]	1.244*** (0.050) [0.000]	0.987*** (0.039) [0.000]	0.505*** (0.061) [0.000]	0.304** (0.114) [0.012]	0.755*** (0.079) [0.000]	0.488*** (0.075) [0.000]	0.403*** (0.049) [0.000]	0.250*** (0.046) [0.000]	0.466*** (0.083) [0.000]	0.497*** (0.041) [0.000]	0.604*** (0.040) [0.000]	0.348*** (0.041) [0.000]	0.811*** (0.057) [0.000]	0.656*** (0.034) [0.000]
Number of Students	20,784	6,860	6,858	7,066	4,052	1,302	1,356	1,395	11,528	3,908	3,895	3,767	36,363	12,042	12,089	12,239
Number of Schools	276	270	266	268	62	61	62	62	145	145	145	145	483	475	473	475
<i>Panel B: 2-Year Exposure</i>																
All students	0.623*** (0.044) [0.000]	0.356*** (0.067) [0.000]	0.953*** (0.056) [0.000]	0.610*** (0.059) [0.000]	0.579*** (0.095) [0.000]	0.305** (0.132) [0.034]	0.729*** (0.113) [0.000]	0.583*** (0.069) [0.000]					0.606*** (0.051) [0.000]	0.358*** (0.079) [0.000]	0.846*** (0.062) [0.000]	0.601*** (0.047) [0.000]
Number of Students	11,150	3,726	3,695	3,729	1,631	505	546	573					12,755	4,203	4,221	4,277
Number of Schools	184	182	178	180	33	32	33	33					217	213	211	213
<i>Panel C: 3-Year Exposure</i>																
All students	0.532*** (0.084) [0.000]	0.238* (0.136) [0.086]	0.941*** (0.099) [0.000]	0.544*** (0.124) [0.000]												
Number of Students	4,053	1,324	1,338	1,367												
Number of Schools	88	87	86	87												

Note: This table reports the program's intent-to-treat (ITT) effects among all students after one year of exposure to PSP (Panel A), two years of exposure (Panel B), and three years of exposure (Panel C). Test performance is measured as the standardized raw total test score, where standardization is performed within each exam wave relative to the control group for the corresponding cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are shown in brackets. Pooled estimates give each cohort equal total weight. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

equation, the average trend in the outcomes over time in each PSP school would have been the same as the average trend in that school’s matched non-PSP school if the PSP had not been implemented in that PSP school. Figures 1, 2 and 3 provide evidence supporting this assumption for Cohorts 1, 2 and 3, respectively: in the seven (school) years before the PSP program started (from 2015-16 to 2021-22), for each cohort changes over time in the Grade 6 exit exam scores are almost identical for the PSP schools and their associated comparison schools.

Table 1 presents the main results of this report for all three cohorts. The first four columns present the results for Cohort 1, which has been implementing the PSP for three years (since September of 2023). Column (5) - (8) present the results for Cohort 2, which has been implementing the PSP for two years (since September of 2024), and columns (9) - (12) present the results for Cohort 3, which has been implementing the PSP for only one year (since September of 2025). Finally, columns (13) - (16) present pooled results that combine cohorts, which will be explained in more detail below. Recall that these estimates are not based on IRT modeling, which was used in the previous reports and will also be applied in the final report. Instead, student performance is based on the percentage of correct answers, normalized using the mean and standard deviation of the students in the comparison schools, calculated for each assessment (test), separately for each grade, cohort and year (and separately for baseline and endline tests within any given year). Even so, the results here for Cohort 1 after two years of exposure and for Cohort 2 after one year of exposure are very similar to the IRT results provided in previous years.

Turning to Cohort 1, after one year of exposure, and combining all three exams, the PSP increased students’ academic skills by 0.90 standard deviations (henceforth s.d.). The subject-specific impacts for Cohort 1 after one year of exposure are 0.49 s.d. for Arabic, 1.24 s.d. for French, and 0.99 s.d. for mathematics. After two years of exposure, the impacts are smaller than the impacts after one year of exposure: 0.62 s.d. for all three subjects combined, 0.36 s.d. for Arabic, 0.95 s.d. for French, and 0.61 s.d. for mathematics. This decline was also seen in the Year 2 final report, yet they are all highly statistically significant and are very high in absolute terms. Using the distribution of impacts over 96 studies in developing countries compiled by Evans and Yuan (2022), the 0.36 s.d. impact for Arabic is slightly above the 80th percentile for reading, the 0.95 s.d. impact for French is above the 99th percentile, and the 0.61 impact for mathematics is at about the 95th percentile. Finally, after three years of exposure, the impacts for Cohort 1 are slightly smaller than the impact after two years of exposure: 0.53 s.d. for all three subjects combined, 0.24 s.d. for Arabic, 0.94 s.d. for French, and 0.54 s.d. for mathematics. Note that the impact for Arabic is statistically significant only at the 10 percent level, which is due to both the smaller estimate and the larger standard error due to the much smaller sample size. All other estimated effects for Cohort 1 are highly statistically significant for all three years. The reasons for this decline are still under investigation and will be discussed in detail in the final report.

Next, turn to the estimated impacts for Cohort 2, which started the PSP program in September of 2024; the results are shown in columns (5) - (8) of Table 1. These results could differ from those of Cohort 1 because Cohort 2 schools are somewhat different from the Cohort 1; for example, as seen in Table 6 (second and fourth columns), the Cohort 2 schools have a slightly higher repetition rate, a slightly higher percentage of students who qualify for the Tayssir program, and are slightly more likely to be in rural areas. After one year of exposure, the

estimated impacts are: 0.51 s.d. for all three subjects combined, 0.30 s.d. for Arabic, 0.76 s.d. for French, and 0.49 s.d. for mathematics. These smaller impacts, compared to the estimates after one year of exposure for Cohort (1), were reported previously in the Year 2 Final Report. The reasons for this lower impact after one year of exposure are still being investigated; two possibilities are that the schools that started the PSP in the second year were not as motivated as those that started in the first year, and that the expansion of the program to a much larger group of schools (2,000, versus 626 in the first year) led to some difficulties in implementing the program. Yet these are still very large effects, and all are highly statistically significant; in terms of the results compiled by Evans and Yuan (2022), the 0.30 s.d. impact for Arabic is slightly below the 75th percentile for reading, the 0.76 s.d. impact for French is at about the 95th percentile, and the 0.49 s.d. impact for mathematics is at about the 95th percentile.

Moving to the results for Cohort 2 after two years of exposure, the estimated impacts are mostly larger than those after one year of exposure. More specifically, the increases in student performance were 0.58 s.d. for all three subjects combined, 0.31 s.d. for Arabic, 0.73 s.d. for French, and 0.58 for mathematics. These results are all statistically significant despite the relatively small sample size. This modest increase in the combined estimate, from 0.51 s.d. to 0.58 s.d., reflects very a small increase in Arabic (from 0.30 s.d. to 0.31 s.d.) and a large increase in mathematics (from 0.49 s.d. to 0.58 s.d.), and a modest decline in French (from 0.76 s.d. to 0.73 s.d.). This further, albeit modest, increase after a second year of exposure is what one would expect. Indeed, these estimated impacts for Cohort 2 after two years of exposure are almost as large as the estimated impacts for Cohort 1 after two years of exposure.

Finally, turning to Cohort 3. Again, These results could differ from those of Cohorts 1 and 2 because Cohort 3 schools are different from those in Cohorts 1 and 2; for example, as seen in Table 6 (fourth and sixth columns), the Cohort 3 schools have a much higher percentage of students who qualify for the Tayssir program, and are much more likely to be in rural areas. The estimated impacts for Cohort 3 in Table 1 after one year of exposure to the program are, except for mathematics, somewhat smaller than the estimated impacts for Cohort 2 after one year of exposure: 0.40 s.d. for all three subjects combined, 0.25 s.d. for Arabic, 0.47 s.d. for French, and 0.50 s.d. for mathematics. Even so, these estimated impacts are all statistically significant and are quite large; in terms of the results compiled by Evans and Yuan (2022), the 0.25 s.d. impact for Arabic is at the 70th percentile for reading, the 0.47 s.d. impact for French is slightly below the 90th percentile, and the 0.50 s.d. impact for mathematics is at about the 95th percentile.

The last four columns (13) - 16) present results that combine cohorts, which produce estimates that are averages over cohorts. Panel A presents the average impact of the PSP program after one year of exposure, averaged over all three cohorts. The estimated impacts are 0.60 s.d. for all three subjects combined, 0.35 s.d. for Arabic, 0.81 s.d. for French, and 0.66 s.d. for mathematics. All of these estimates are highly significant and are very large effects. Panel B presents the average impact of the PSP program after two years of exposure, averaged over Cohorts 1 and 2. The estimated impacts are very similar to the averages after one year of exposure for all three cohorts: 0.61 s.d. for all three subjects combined, 0.36 s.d. for Arabic, 0.85 s.d. for French, and 0.60 s.d. for mathematics. Again, all of these estimates are highly significant and are very large effects.

3 Stable Sample Analysis

The changes over time for the Cohort 1 and 2 results in Table 1 could be misleading because the samples change, becoming smaller from one year to the next because some comparison schools participate in the PSP in later years, and so those schools, and their paired PSP schools, must be dropped from the analysis. To get a clearer understanding of how the estimated impacts change over time, this section presents re-estimated results of the impact of the PSP for Cohorts 1 and 2 using only the schools that are still in the sample in the third year of data collection. This removes potential concerns that the changes in the effectiveness of PSP are due to changes in the composition of the students taking the exam. For example, a drop in the ITT effect may occur because the pairs of schools that are dropped from the sample over time are pairs where the differences in the performance of the PSP and the comparison school were unusually large. This would artificially deflate the estimated ITT effects over time.

We present these results in Table 2.

Table 2: *Students Performance After 3 Years with Stable Samples*

	1-Year Exposure				2-Year Exposure				3-Year Exposure			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)
Panel A: Cohort 1 (2023/24)												
All students	0.964*** (0.057) [0.000]	0.514*** (0.086) [0.000]	1.506*** (0.107) [0.000]	0.888*** (0.081) [0.000]	0.566*** (0.059) [0.000]	0.299*** (0.104) [0.006]	0.880*** (0.072) [0.000]	0.538*** (0.085) [0.000]	0.441*** (0.069) [0.000]	0.167** (0.063) [0.012]	0.729*** (0.078) [0.000]	0.530*** (0.121) [0.000]
Number of Students	3,711	1,213	1,204	1,294	3,920	1,283	1,299	1,338	3,932	1,286	1,300	1,346
Number of Schools	86	83	83	85	86	85	84	85	86	85	84	86
Panel B: Cohort 2 (2024/25)												
All students	0.514*** (0.101) [0.000]	0.304* (0.151) [0.064]	0.671*** (0.140) [0.000]	0.435*** (0.107) [0.001]	0.474*** (0.073) [0.000]	0.231** (0.103) [0.041]	0.552*** (0.086) [0.000]	0.569*** (0.067) [0.000]				
Number of Students	1,486	464	500	522	1,577	488	533	556				
Number of Schools	30	29	30	30	32	31	32	32				

Note: This table reports on the program’s intent-to-treat (ITT) effects among all students after for Cohort 1 (Panel A) and Cohort 2 (Panel B). The samples have been modified to become stable to the attrition observed at the end of 3 years of data collection. The outcome of student performance is measured as the standardized test performance in each exam wave relative to the control group for each sample cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are reported in brackets. Pooled estimates are based on cohort-reweighted samples. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The results in Table 2 show patterns that are similar to those in Table 1, which indicates that the patterns in Table 1 are not due to changes in the samples of schools and students over time. More specifically, for Cohort 1 there is still a sizable drop in the impact of the PSP after two years of exposure, relative to one year of exposure, and there is a smaller drop after three years of exposure (relative to two years of exposure) than the drop observed after two years of exposure (relative to one year of exposure). For Cohort 2, the impact after two years of exposure is slightly smaller than the impact after one year and thus does not show the modest rise seen in Table 1 (for all three subjects combined, the increase in Table 1 is from 0.51 s.d to 0.58 s.d., while the decrease in Table 2 is from 0.51 s.d. to 0.47 s.d.) . Yet for two

of the three subjects the direction of change is the same: the estimated impact declined in French and increased in mathematics, and only the Arabic result reverses sign (from a tiny increase in Table 1 to a sizable decrease in Table 2). We conclude that, overall, the changes in estimated impacts over time for Cohorts 1 and 2 in Table 1 cannot be explained by changes in the sample over time.

4 Heterogeneity Analysis

We assess heterogeneity of the results along two dimensions. First, we assess differences in the impact of PSP between boys and girls. Second, we assess differences in the impact of PSP between the top quartile and bottom quartile of baseline performing students.

Boys vs. Girls. Table 3 presents the same estimates as in Table 1, but separately for the female and male students in our sample. Overall, when combining the three subjects, the estimated impacts are larger for girls than for boys, but the differences are not particularly large. By subject, the impacts on all three subjects (Arabic, French and mathematics) are almost always larger for girls than for boys, the main exception being that while the impacts on French skills for Cohort 2 are slightly higher for boys than for girls. But none of these differences is very large. We conclude that the PSP program is similarly effective at increasing the academic skills of boys and girls.

Baseline Performance. Table 4 also presents the same estimates as in Table 1, but it presents separate results for the bottom quartile and top quartile of test scores as measured at baseline. This is done to see whether the program is effective at raising students' skills at both tails of the distribution of initial performance. We ranked students in terms of their baseline performance and separated them into quartiles along this dimension. Then we estimate the impact of the PSP separately for the top quartile of baseline performing students and the bottom quartile of baseline performing students.

After one year of exposure, Cohort 1 students in the bottom quartile increase their performance, averaged over all three tests, by 0.85 s.d. For the top quartile the increase is slightly smaller, at 0.82 s.d. After two years of exposure for Cohort 1 students, the impact on the bottom quartile is – as expected given the results in Table 1 – much smaller, at 0.59 s.d., while the impact on the top quartile also declines, to 0.54 s.d. Finally, after three years of exposure, the impacts drop further for both quartiles: 0.52 s.d. for the bottom quartile and 0.46 s.d. for the top quartile. Overall, the PSP increased student performance for students in both the top and bottom quartiles, but the impact is slightly larger for the bottom quartile of students.

Turning to Cohort 2, and again focusing on the overall scores that average over the three subjects, after one year of exposure the bottom quartile's scores increase by 0.53 s.d. while the top quartile's increase is noticeably smaller (although still large), at 0.37 s.d. After a second year of exposure, the bottom quartile's increase is larger, at 0.63 s.d., while the top quartile's performance is much higher, at 0.50 s.d. Finally, for Cohort 3 after one year of exposure, the overall impact of the PSP is an increase of 0.36 s.d. for the bottom cohort and 0.34 s.d. for the top cohort. These results again show that the PSP increased students' skills for both cohorts, but slightly more for the bottom cohort than for the top cohort.

Table 3: Heterogeneity in Students' Performance by Gender

	Cohort 1 (2023/24)				Cohort 2 (2024/25)				Cohort 3 (2025/26)				Cohorts Pooled			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
<i>Panel A: 1-Year Exposure</i>																
Female	0.939*** (0.032) [0.000]	0.475*** (0.045) [0.000]	1.280*** (0.051) [0.000]	1.024*** (0.044) [0.000]	0.538*** (0.072) [0.000]	0.408*** (0.116) [0.001]	0.740*** (0.084) [0.000]	0.537*** (0.083) [0.000]	0.448*** (0.041) [0.000]	0.272*** (0.053) [0.000]	0.509*** (0.069) [0.000]	0.515*** (0.048) [0.000]	0.642*** (0.032) [0.000]	0.383*** (0.043) [0.000]	0.835*** (0.044) [0.000]	0.692*** (0.039) [0.000]
Number of Students	10,271	3,345	3,420	3,388	1,990	616	677	682	5,559	1,834	1,858	1,805	17,820	5,795	5,955	5,875
Number of Schools	276	270	265	268	62	60	62	62	145	145	145	145	483	475	472	475
Male	0.865*** (0.033) [0.000]	0.500*** (0.050) [0.000]	1.185*** (0.060) [0.000]	0.942*** (0.047) [0.000]	0.478*** (0.061) [0.000]	0.224* (0.119) [0.071]	0.753*** (0.082) [0.000]	0.446*** (0.072) [0.000]	0.362*** (0.050) [0.000]	0.215*** (0.046) [0.000]	0.433*** (0.084) [0.000]	0.441*** (0.044) [0.000]	0.567*** (0.041) [0.000]	0.307*** (0.045) [0.000]	0.775*** (0.061) [0.000]	0.612*** (0.036) [0.000]
Number of Students	10,511	3,437	3,356	3,605	2,062	671	666	710	5,966	2,013	1,987	1,930	18,539	6,121	6,009	6,245
Number of Schools	276	270	265	268	62	60	62	62	145	145	145	145	483	475	472	475
<i>Panel B: 2-Year Exposure</i>																
Female	0.687*** (0.047) [0.000]	0.370*** (0.071) [0.000]	1.000*** (0.065) [0.000]	0.627*** (0.069) [0.000]	0.672*** (0.115) [0.000]	0.452*** (0.125) [0.003]	0.712*** (0.120) [0.000]	0.634*** (0.103) [0.000]					0.679*** (0.059) [0.000]	0.465*** (0.075) [0.000]	0.859*** (0.070) [0.000]	0.635*** (0.063) [0.000]
Number of Students	5,497	1,794	1,869	1,747	781	214	270	272					6,278	2,008	2,139	2,019
Number of Schools	184	182	178	180	33	31	33	33					217	213	211	213
Male	0.561*** (0.046) [0.000]	0.305*** (0.078) [0.000]	0.875*** (0.061) [0.000]	0.607*** (0.069) [0.000]	0.527*** (0.089) [0.000]	0.164 (0.143) [0.267]	0.720*** (0.132) [0.000]	0.526*** (0.063) [0.000]					0.547*** (0.050) [0.000]	0.224** (0.087) [0.011]	0.801*** (0.072) [0.000]	0.573*** (0.051) [0.000]
Number of Students	5,623	1,863	1,760	1,894	831	263	260	285					6,454	2,126	2,020	2,179
Number of Schools	184	181	178	180	33	31	33	33					217	212	211	213
<i>Panel C: 3-Year Exposure</i>																
Female	0.616*** (0.091) [0.000]	0.334** (0.161) [0.044]	1.039*** (0.124) [0.000]	0.575*** (0.093) [0.000]												
Number of Students	1,970	603	669	627												
Number of Schools	88	87	86	86												
Male	0.465*** (0.089) [0.000]	0.126 (0.111) [0.261]	0.808*** (0.081) [0.000]	0.443*** (0.146) [0.004]												
Number of Students	2,032	648	607	682												
Number of Schools	88	87	85	86												

Note: This table reports on the program's intent-to-treat (ITT) effects between female and male students after 1-year of exposure to PSP (Panel A), 2-years of exposure to PSP (Panel B), and 3-years of exposure to PSP (Panel C). The outcome of student performance is measured as the standardized test performance in each exam wave relative to the control group for each sample cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are reported in brackets. Pooled estimates are based on cohort-reweighted samples. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

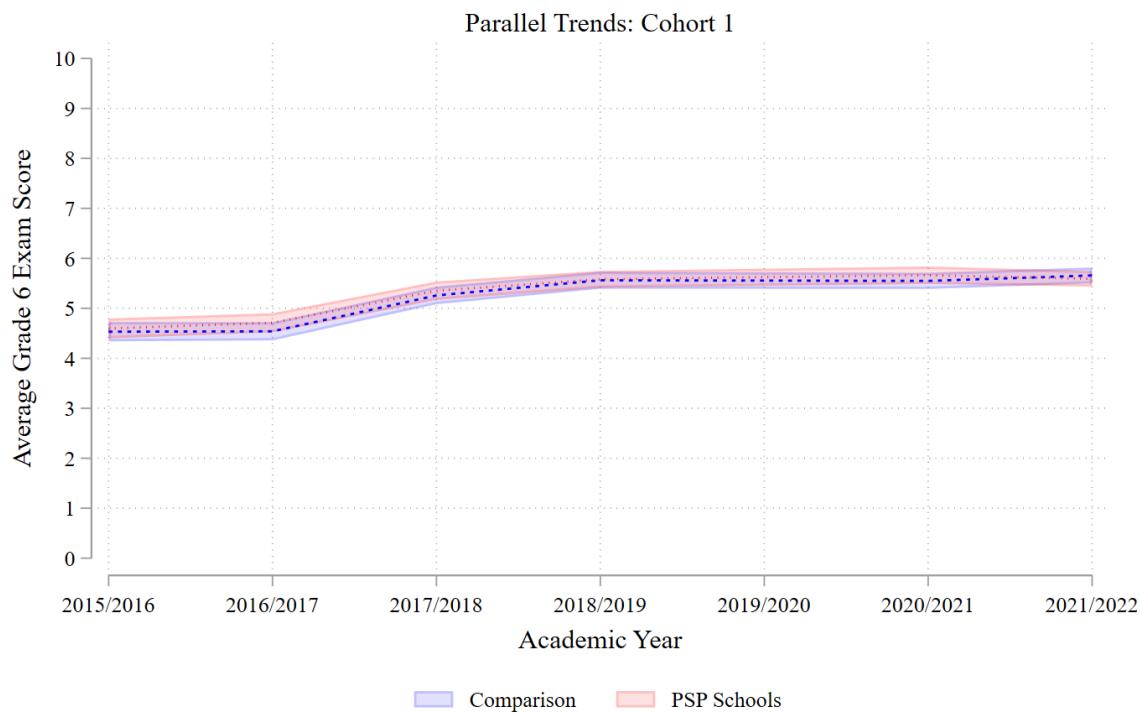
Table 4: Heterogeneity in Students' Performance by Baseline Performance

	Cohort 1 (2023/24)				Cohort 2 (2024/25)				Cohort 3 (2025/26)				Cohorts Pooled			
	Overall (1)	Arabic (2)	French (3)	Math (4)	Overall (5)	Arabic (6)	French (7)	Math (8)	Overall (9)	Arabic (10)	French (11)	Math (12)	Overall (13)	Arabic (14)	French (15)	Math (16)
<i>Panel A: 1-Year Exposure</i>																
Bottom quartile	0.854*** (0.047) [0.000]	0.565*** (0.064) [0.000]	1.148*** (0.079) [0.000]	0.872*** (0.057) [0.000]	0.529*** (0.093) [0.000]	0.373*** (0.134) [0.009]	0.577*** (0.125) [0.000]	0.436*** (0.110) [0.000]	0.359*** (0.073) [0.000]	0.213*** (0.069) [0.003]	0.395*** (0.117) [0.001]	0.398*** (0.070) [0.000]	0.595*** (0.058) [0.000]	0.398*** (0.055) [0.000]	0.734*** (0.088) [0.000]	0.574*** (0.051) [0.000]
Number of Students	6,573	1,914	2,269	2,200	1,162	323	397	418	3,161	951	1,011	967	10,896	3,177	3,665	3,596
Number of Schools	276	250	249	265	62	56	57	59	145	135	132	134	483	441	438	458
Top quartile	0.816*** (0.047) [0.000]	0.404*** (0.082) [0.000]	0.671*** (0.123) [0.000]	1.245*** (0.092) [0.000]	0.373*** (0.084) [0.000]	0.322** (0.150) [0.041]	0.639*** (0.098) [0.000]	0.457*** (0.135) [0.002]	0.339*** (0.049) [0.000]	0.168** (0.071) [0.020]	0.434*** (0.089) [0.000]	0.324*** (0.079) [0.000]	0.469*** (0.040) [0.000]	0.275*** (0.065) [0.000]	0.536*** (0.061) [0.000]	0.567*** (0.072) [0.000]
Number of Students	2,980	959	532	649	783	260	192	224	2,386	805	640	602	6,149	2,024	1,364	1,476
Number of Schools	271	213	138	189	61	51	48	54	144	131	110	126	476	395	296	369
<i>Panel B: 2-Year Exposure</i>																
Bottom quartile	0.590*** (0.059) [0.000]	0.372*** (0.083) [0.000]	0.796*** (0.101) [0.000]	0.607*** (0.066) [0.000]	0.631*** (0.154) [0.001]	0.304** (0.126) [0.033]	0.568** (0.229) [0.025]	0.729*** (0.174) [0.001]					0.618*** (0.078) [0.000]	0.348*** (0.071) [0.000]	0.691*** (0.113) [0.000]	0.667*** (0.096) [0.000]
Number of Students	3,258	1,039	1,178	1,086	468	120	139	149					3,726	1,159	1,317	1,235
Number of Schools	183	169	167	173	33	25	29	28					216	194	196	201
Top quartile	0.542*** (0.054) [0.000]	0.307** (0.122) [0.014]	0.790*** (0.136) [0.000]	0.709*** (0.141) [0.000]	0.496*** (0.128) [0.001]	0.224 (0.338) [0.525]	0.767*** (0.202) [0.002]	0.147 (0.191) [0.458]					0.519*** (0.071) [0.000]	0.313** (0.143) [0.031]	0.773*** (0.138) [0.000]	0.395*** (0.120) [0.001]
Number of Students	1,855	503	383	450	292	71	94	77					2,147	574	477	527
Number of Schools	182	136	106	133	33	19	27	20					215	155	133	153
<i>Panel C: 3-Year Exposure</i>																
Bottom quartile	0.517*** (0.119) [0.000]	0.222** (0.110) [0.050]	0.770*** (0.159) [0.000]	0.589*** (0.156) [0.000]												
Number of Students	1,251	392	376	438												
Number of Schools	88	79	78	85												
Top quartile	0.463*** (0.144) [0.002]	0.336** (0.148) [0.029]	0.779*** (0.188) [0.000]	0.370 (0.261) [0.166]												
Number of Students	781	203	195	159												
Number of Schools	88	64	54	53												

Note: This table reports on the program's intent-to-treat (ITT) effects between the top quartile and bottom quartile of students based on their baseline test performance after 1-year of exposure to PSP (Panel A), 2-years of exposure to PSP (Panel B), and 3-years of exposure to PSP (Panel C). The outcome of student performance is measured as the standardized test performance in each exam wave relative to the control group for each sample cohort. Each subject level exams (Arabic, French, and Math) is reported alongside a composite score stacking the three sub-samples. Standard errors are clustered at the matched-pair level and shown in parentheses. P-values are reported in brackets. Pooled estimates are based on cohort-reweighted samples. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

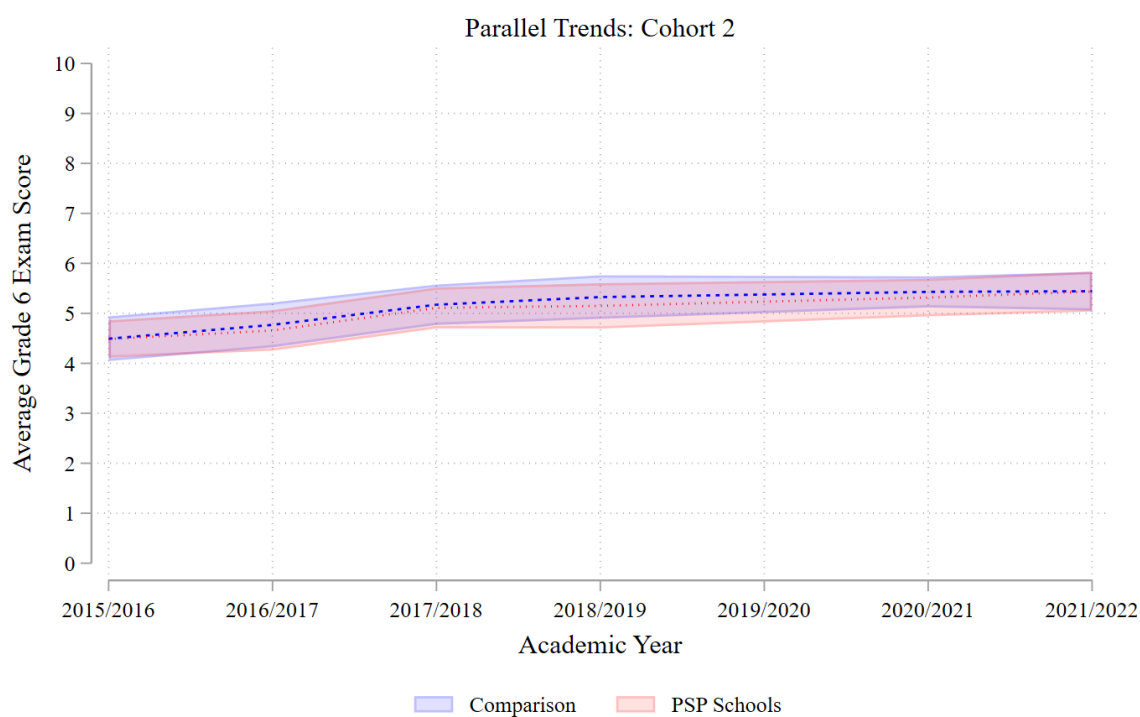
5 Parallel Trends Figures and Characteristics of the Three Cohorts

Figure 1: *Parallel Trends of Test Scores Across Treatment Assignment by Cohort 1 School*



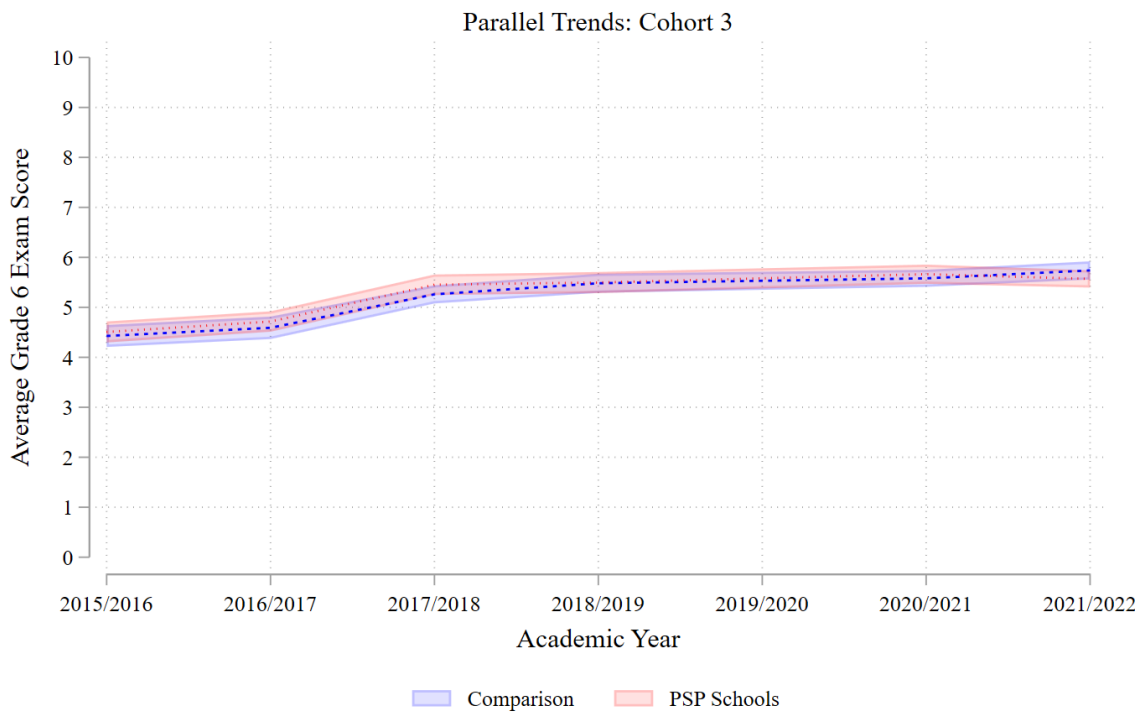
Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents the yearly average end-of-primary exam scores over a period of seven years before the program launched. Cohort 1 represents the original 300 selected schools (150 PSP, 150 comparison).

Figure 2: *Parallel Trends of Test Scores Across Treatment Assignment by Cohort 2 School*



Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents the yearly average end-of-primary exam scores over a period of seven years before the program launched. Cohort 2 represents the original 70 selected schools (35 PSP, 35 comparison).

Figure 3: *Parallel Trends of Test Scores Across Treatment Assignment by Cohort 2 School*



Note: This figure confirms that the program and comparison schools exhibited similar levels and similar trends in exam scores before the program started (2022/23). It presents the yearly average end-of-primary exam scores over a period of seven years before the program launched. Cohort 3 represents the original 270 selected schools (135 PSP, 135 comparison).

References

- Belloni, A., V. Chernozhukov, and C. Hansen (2011). Inference for high-dimensional sparse econometric models. *arXiv*.
- de Chaisemartin, C. and J. Ramirez-Cuellar (2024). At what level should one cluster standard errors in paired and small-strata experiments? *American Economic Journal: Applied Economics* 16, 193–212.
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Table 5: *Internal Validity of Schools in Study Sample*

	Comparison - Cohort 1	PSP - Cohort 1	Difference	Comparison - Cohort 2	PSP - Cohort 2	Difference	Comparison - Cohort 3	PSP - Cohort 3	Difference
Grade 6 Final Exam Score (avg. per classroom, 2021/2022)	5.651	5.595	0.013	5.443	5.436	0.323	5.739	5.569	-0.162
	0.976	0.994	0.899	1.166	1.210	0.237	1.064	1.026	0.171
Grade 6 Final Exam Score (avg. per classroom, 2020/2021)	5.549	5.671	0.173	5.429	5.315	0.158	5.582	5.663	0.077
	0.968	1.140	0.104	0.938	1.134	0.503	1.007	1.143	0.536
Grade 6 Final Exam Score (avg. per classroom, 2018/2019)	5.569	5.575	0.061	5.325	5.149	-0.195	5.482	5.496	0.051
	1.026	1.085	0.574	1.314	1.370	0.498	1.127	1.250	0.707
Grade 6 Final Exam Score (avg. per classroom, 2017/2018)	5.275	5.342	0.174	5.173	5.108	-0.123	5.260	5.452	0.255
	1.053	1.197	0.118	1.216	1.236	0.604	1.079	1.221	0.038
Grade 6 Final Exam Score (avg. per classroom, 2016/2017)	4.564	4.699	0.242	4.769	4.658	-0.012	4.590	4.716	0.197
	1.119	1.226	0.043	1.355	1.221	0.963	1.332	1.204	0.178
Grade 6 Final Exam Score (avg. per classroom, 2015/2016)	4.535	4.601	0.142	4.492	4.486	-0.088	4.427	4.507	0.107
	1.177	1.273	0.286	1.347	1.129	0.724	1.303	1.248	0.454
Total Enrollment (2021/2022)	508.420	497.727	-14.910	497.400	484.114	-28.002	450.704	448.281	-7.237
	332.208	312.852	0.699	326.525	332.736	0.776	370.485	330.399	0.865
Female Enrollment (% , 2021/2022)	0.482	0.483	-0.000	0.485	0.478	-0.009	0.481	0.481	0.001
	0.032	0.034	0.963	0.037	0.029	0.338	0.042	0.032	0.873
Students Repeating (% , 2021/2022)	0.080	0.070	-0.011	0.083	0.086	-0.006	0.085	0.078	-0.006
	0.039	0.033	0.013	0.039	0.041	0.514	0.045	0.038	0.215
Student Dropout (% , 2021/2022)	0.013	0.010	-0.003	0.010	0.009	-0.002	0.011	0.011	0.000
	0.015	0.012	0.042	0.012	0.010	0.582	0.016	0.013	0.971
Students Qualifying for Tayssir (% , 2021/2022)	0.201	0.208	-0.005	0.244	0.217	-0.056	0.370	0.345	-0.030
	0.307	0.290	0.886	0.331	0.291	0.480	0.329	0.327	0.435
Students Qualifying for Social Security (% , 2021/2022)	0.203	0.209	-0.005	0.244	0.217	-0.055	0.372	0.346	-0.030
	0.311	0.291	0.889	0.331	0.292	0.484	0.331	0.327	0.431
Urban	0.680	0.640	-0.024	0.629	0.629	0.060	0.400	0.430	0.030
	0.468	0.482	0.651	0.490	0.490	0.650	0.492	0.497	0.607
Number of Teachers	18.620	18.473	-0.330	18.800	18.743	-0.670	16.711	17.141	0.370
	10.137	10.073	0.789	9.737	9.784	0.818	11.542	10.842	0.788
School's Inspector is Trained	0.820	0.833	0.072	0.171	0.143	-0.033	0.400	0.370	-0.051
	0.385	0.374	0.063	0.382	0.355	0.716	0.492	0.485	0.395
School Type: Community	0.007	0.020	0.019	0.029	0.029	-0.009	0.007	0.022	0.018
	0.082	0.140	0.194	0.169	0.169	0.836	0.086	0.148	0.255
School Type: Central/Sector	0.280	0.287	-0.019	0.286	0.286	-0.091	0.489	0.452	-0.044
	0.451	0.454	0.707	0.458	0.458	0.432	0.502	0.500	0.476
Share of Multi-level classrooms (% , 2021/2022)	0.045	0.057	0.007	0.055	0.078	0.033	0.134	0.125	-0.014
	0.162	0.180	0.719	0.182	0.183	0.500	0.293	0.276	0.689
Share of Multi-level Grades (% , 2021/2022)	0.027	0.036	0.005	0.033	0.029	-0.006	0.111	0.095	-0.022
	0.162	0.173	0.775	0.171	0.169	0.900	0.294	0.261	0.508
# of Schools	150	150	.	35	35	.	135	135	.
Joint Wald test p-value	.	.	0.004	.	.	0.107	.	.	0.256

Note: This table compares the school level baseline covariates across PSP and comparison schools for each cohort in our sample. For each group we present both the mean and the standard deviation. The difference is the difference in means controlling for regional fixed effects. Below the difference is the heteroskedastic corrected p-value. At the bottom of the table, we report the sample size of schools for each group. And add the bottom of the differences columns we report a joint Wald test p-value for the joint significance that all reported covariates are jointly different across groups.

Table 6: *External Validity of Schools in Study Sample*

	Non-Study	Cohort 1	Difference	Cohort 2	Difference	Cohort 3	Difference	All Cohorts	Difference
Grade 6 Final Exam Score (avg. per classroom, 2021/2022)	5.744	5.623	-0.001	5.440	-0.000	5.654	0.000	5.654	-0.001
	1.076	0.984	0.230	1.179	0.368	1.047	0.313	1.020	0.107
Grade 6 Final Exam Score (avg. per classroom, 2020/2021)	5.722	5.610	-0.001	5.372	-0.000	5.622	0.000	5.613	-0.002
	1.146	1.057	0.080	1.035	0.155	1.076	0.850	1.067	0.022
Grade 6 Final Exam Score (avg. per classroom, 2018/2019)	5.469	5.572	0.001	5.237	-0.000	5.489	0.001	5.540	0.001
	1.224	1.054	0.026	1.336	0.745	1.188	0.110	1.122	0.072
Grade 6 Final Exam Score (avg. per classroom, 2017/2018)	5.279	5.309	0.000	5.141	0.000	5.356	0.001	5.354	0.001
	1.273	1.126	0.393	1.218	0.942	1.154	0.003	1.149	0.065
Grade 6 Final Exam Score (avg. per classroom, 2016/2017)	4.516	4.631	0.001	4.714	0.000	4.653	0.001	4.671	0.002
	1.386	1.174	0.105	1.282	0.010	1.269	0.052	1.231	0.001
Grade 6 Final Exam Score (avg. per classroom, 2015/2016)	4.371	4.568	0.001	4.489	0.000	4.467	0.001	4.550	0.003
	1.407	1.225	0.001	1.234	0.038	1.274	0.076	1.257	0.000
Total Enrollment (2021/2022)	194.937	503.073	0.000	490.757	0.000	449.493	0.000	475.618	0.000
	253.353	322.179	0.000	327.316	0.000	350.363	0.000	333.559	0.000
Female Enrollment (% , 2021/2022)	0.480	0.482	0.006	0.482	0.000	0.481	0.002	0.482	0.008
	0.073	0.033	0.189	0.033	0.809	0.037	0.714	0.036	0.299
Students Repeating (% , 2021/2022)	0.089	0.075	-0.032	0.084	-0.002	0.082	-0.010	0.078	-0.055
	0.062	0.037	0.000	0.039	0.480	0.042	0.092	0.038	0.000
Student Dropout (% , 2021/2022)	0.010	0.011	-0.005	0.010	-0.013	0.011	-0.000	0.011	-0.007
	0.021	0.014	0.801	0.011	0.013	0.015	0.989	0.013	0.839
Students Qualifying for Tayssir (% , 2021/2022)	0.565	0.205	-0.045	0.230	-0.008	0.358	-0.012	0.265	-0.079
	0.290	0.298	0.000	0.310	0.000	0.328	0.000	0.320	0.000
Students Qualifying for Social Security (% , 2021/2022)	0.569	0.206	-0.044	0.230	-0.008	0.359	-0.012	0.267	-0.078
	0.293	0.300	0.000	0.310	0.000	0.329	0.000	0.322	0.000
Urban	0.147	0.660	0.043	0.629	0.008	0.415	0.010	0.568	0.071
	0.355	0.475	0.000	0.487	0.000	0.494	0.000	0.496	0.000
Number of Teachers	7.895	18.547	0.001	18.771	0.000	16.926	0.001	17.787	0.003
	8.335	10.088	0.000	9.690	0.000	11.179	0.000	10.522	0.000
School's Inspector is Trained	0.075	0.827	0.098	0.157	0.001	0.385	0.008	0.552	0.141
	0.263	0.379	0.000	0.367	0.452	0.488	0.008	0.498	0.000
School Type: Community	0.013	0.013	-0.000	0.029	-0.002	0.015	-0.003	0.014	0.001
	0.112	0.115	0.996	0.168	0.000	0.121	0.463	0.116	0.935
School Type: Central/Sector	0.196	0.283	0.002	0.286	-0.000	0.470	0.014	0.348	0.024
	0.397	0.451	0.175	0.455	0.386	0.500	0.000	0.477	0.000
Share of Multi-level classrooms (% , 2021/2022)	0.541	0.051	-0.022	0.067	-0.004	0.129	-0.011	0.098	-0.050
	0.452	0.171	0.000	0.181	0.000	0.284	0.000	0.245	0.000
Share of Multi-level Grades (% , 2021/2022)	0.517	0.031	-0.020	0.031	-0.004	0.103	-0.010	0.073	-0.045
	0.475	0.167	0.000	0.169	0.000	0.277	0.000	0.240	0.000
# of Schools	20566	300	.	70	.	270	.	640	.
Joint Wald test p-value	.	.	0.000	.	0.033	.	0.000	.	0.000

Note: This table compares the school level baseline covariates across non-study schools and the schools selected for each cohort sample. For each group we present both the mean and the standard deviation. The difference is the difference in means controlling for regional fixed effects. Below the difference is the heteroskedastic corrected p-value. At the bottom of the table, we report the sample size of schools for each group. And add the bottom of the differences columns we report a joint Wald test p-value for the joint significance that all reported covariates are jointly different across groups.